Project Topics

Projects fall into two categories.

A) Application Projects implement a complete biometric recognition pipeline on public data. They apply the methods covered in the course from end to end.

B) Research Projects address open problems identified in the recent literature (2024–2026) and carry a heavier workload. Each topic defines a minimum scope that must be completed within the course. Research projects may be continued after the course with the instructor, with the aim of publication.

The choice of category does not affect grading; projects are graded on rigor, not on whether the results are positive.

Each topic states its **evaluation** (metrics, data split and main curve) and the kind of **reference result** expected. These complete the evaluation rules of the project description.

Rules for application projects:

- You may use open-source implementations for any stage of the pipeline, provided you declare them and can explain how they work. At least one stage must be your own implementation.

Rules for research projects:

- Each research topic is assigned to one team only. If several teams request the same topic, the instructor decides.
- In the design document, research teams must confirm from the recent literature that their problem is still open. They must also name the paper that serves as their reference result.

A) Application Projects

A1. Fingerprint Verification: Minutiae vs. Deep Features

Problem. Minutiae matching has been the backbone of fingerprint recognition for decades, and deep learned representations now compete with it. The two approaches capture different information and fail in different situations.

Question. How do a minutiae-based matcher and a deep embedding compare on the same data, and where does each one fail?

Scope.

- Use one database of the Fingerprint Verification Competition (FVC2002 or FVC2004), e.g., DB1.
- Build a classical pipeline: enhancement, minutiae extraction and minutiae matching. Existing tools (e.g., NIST NBIS or SourceAFIS) may be used for extraction and matching.
- Compare it with a pretrained or fine-tuned convolutional neural network (CNN) embedding.
- Analyze the failure cases of both approaches.

Evaluation.

- Metrics: the detection error tradeoff (DET) curve, the equal error rate (EER), and the false non-match rate (FNMR) at a fixed false match rate (FMR).

- Data split: test on set A of the chosen database, following the FVC protocol. Any data used to fine-tune the CNN or to choose thresholds must come from other fingers, e.g., set B or another FVC database.
- Main curve: DET.

Reference result. A published result on the same database and protocol, e.g., the official FVC results.

A2. Face Recognition Under Image Degradations

Problem. Face recognition reaches near-perfect accuracy on good-quality images. Real-world images, however, are often small, blurred, partially occluded or poorly lit.

Question. How quickly does performance degrade under each type of degradation, and are classical and deep methods affected differently?

Scope.

- Use Labeled Faces in the Wild (LFW) in two setups: verification and closed-set identification.
- Compare a classical baseline, either local binary patterns (LBP) or Eigenfaces, with a pretrained deep embedding.
- Apply at least two of the following degradations at increasing severity: low resolution, blur, occlusion and illumination change.

Evaluation.

- Metrics: FNMR at a fixed FMR for verification; rank-1 accuracy and the cumulative match characteristic (CMC) curve for identification.
- Data split: use the ten folds of the LFW verification protocol, which are identity-disjoint. Fit anything that is learned (e.g., the Eigenfaces subspace) and choose thresholds on the folds not used for testing. The standard protocol has only 3,000 impostor pairs; you may form additional impostor pairs within each fold. For identification, fit on identities that are not in the gallery or the probes, and define how you build the gallery and the probes; only identities with at least two images can serve as probes. State the training data of the pretrained model and whether it may contain LFW identities.
- Main curves: DET for verification, CMC for identification.

Reference result. A published result of your pretrained embedding on the standard LFW verification protocol with undegraded images. Such results are usually reported as verification accuracy, so report accuracy as well for this comparison.

A3. An Iris Recognition Pipeline

Problem. Iris recognition is among the most accurate biometric modalities. Its accuracy, however, depends heavily on segmentation, which becomes difficult under poor illumination.

Question. Which stage of the pipeline limits performance, and how much does reduced illumination affect each stage?

Scope.

- Use one of the following datasets:
  - the Interval subset of the CASIA-Iris v4 database of the Institute of Automation, Chinese Academy of Sciences;

    - the IIT Delhi (IITD) iris database.
- Implement the full pipeline: segmentation, rubber-sheet normalization, a Gabor-based IrisCode, and masked Hamming distance with rotation compensation.
- Locate the limiting stage by running the pipeline twice: once with your own segmentation and once with reference segmentation. The difference in performance is the cost of your segmentation.
    - Reference masks: the IRISSEG-EP ground truth, which covers both datasets and is available on request from its authors. If it does not arrive in time, mark the pupil and iris circles (and, if possible, the eyelid boundaries) by hand for all images of about 20 subjects and run the comparison on this subset.
- Simulate reduced illumination at increasing severity (darkening with added sensor noise) and repeat both runs at each level. Because the reference masks remain valid for the darkened images, this separates segmentation failures from the loss of iris texture. Darkening simulates underexposure only; it does not model the pupil's response to light, so state this in your limitations.
- Inspect segmentation failures.

Evaluation.

- Metrics: DET curve, EER and FNMR at a fixed FMR for each condition, together with segmentation accuracy against the reference masks (e.g., F1 score).
- Data split: choose any tuned parameters (filter settings, segmentation parameters, thresholds) on subjects that are not in the test set. The left and right eyes of one person belong to the same subject.
- Main curve: DET.

Reference result. A published result of a Daugman-style IrisCode pipeline on the same database.

A4. Speaker Verification: Classical vs. Deep

Problem. Speaker verification has moved from statistical models of hand-crafted features to deep embeddings. Short and noisy utterances remain challenging for both.

Question. How do classical and deep approaches compare, and how do utterance length and noise affect each?

Scope.

- Use VoxCeleb1 with the VoxCeleb1-O test protocol.
- Compare two approaches:
    - a classical baseline: mel-frequency cepstral coefficients (MFCC) with a Gaussian mixture model–universal background model (GMM-UBM);
    - a deep model: a pretrained ECAPA-TDNN (Emphasized Channel Attention, Propagation and Aggregation in Time Delay Neural Network).
- Evaluate on truncated utterances and on utterances with added noise at several signal-to-noise ratio (SNR) levels. Take the noise from a public noise corpus (e.g., MUSAN).
- Pretrained ECAPA-TDNN models are often trained with noise and reverberation augmentation. Check the model card, use noise files the model was not trained with if possible, and take the augmentation into account when you compare noise robustness with the GMM-UBM.

Evaluation.

- Metrics: DET curve and EER for each condition.
- Data split: train the UBM on the VoxCeleb1 development speakers only. Check that the training data of the pretrained model excludes the VoxCeleb1-O test speakers.
- Main curve: DET.

Reference result. The published EER of your pretrained ECAPA-TDNN on VoxCeleb1-O, and a published GMM-UBM result on VoxCeleb1.

A5. Keystroke Dynamics: Password-Typing Verification Across Sessions

Problem. Typing rhythm can add a behavioral check to a password without extra hardware. A person's typing changes from day to day, however, so results measured within a single session can be optimistic.

Question. How well do simple and learned detectors separate users who type the same password, and how much does performance drop across sessions?

Scope.

- Use the Carnegie Mellon University (CMU) Keystroke Dynamics Benchmark (51 subjects, 8 sessions). All subjects type the same fixed password, so this is static (password-based) authentication, not continuous authentication over free text.
- Compare distance-based detectors (Manhattan, scaled Manhattan, Mahalanobis) with one learned model.
- Evaluate within-session and cross-session conditions separately.
- Optionally, fuse the scores of several consecutive password entries and measure how much this improves the decision.

Evaluation.

- Metrics: DET curve and EER for each condition.
- Data split: train each user's detector on that user's earlier samples and test on later ones; the other subjects serve as impostors. If a detector is trained with impostor samples, those impostor subjects must not be reused as test impostors. A learned model shared across users must be trained on users who are not used for evaluation.
- Main curve: DET.

Reference result. The results of the original benchmark study (Killourhy & Maxion, 2009) for the same detectors.

B) Research Projects

B1. How Many Queries Does It Take? Decision-Only Leakage in Biometric Verification

Problem. A verification system can leak its enrolled templates through its own responses. Protection schemes rarely treat the number of allowed queries, or the granularity of the returned score, as security parameters.

Question. How many queries does an attacker need to get a forged template accepted, and to recover the target's identity? How does this number depend on what the system reveals: a float score, a quantized score, or only accept/reject?

Scope.

- Build a verifier on a public pretrained face matcher, using LFW.
- Threat model: the attacker can submit embeddings directly to the comparator (template injection) and observes only the system's response. State this in the design document.
- Implement the attack in embedding space:
  - for float, 8-bit and 4-bit scores, a hill-climbing attack;
  - for 1-bit (accept/reject only), a decision-based attack: first reach acceptance, then move toward the target by probing the decision boundary. Hill climbing cannot be used here because it receives no signal before the first acceptance.
- Define the score quantization explicitly (score range and level boundaries). If a quantized score gives no signal from the starting points, treat that case like the 1-bit case.
- Draw starting points from a pool of face embeddings whose identities are disjoint from the targets.
- Vary two parameters:
  - the oracle granularity: float, 8-bit, 4-bit and 1-bit (this sweep also serves as the evaluation of score coarsening as a defense);
  - the threshold: FMR = 1% and 0.1%.
- Measure the zero-effort baseline: submitting embeddings from the pool until one is accepted. Its expected cost is about 1/FMR queries (about 100 at FMR = 1%), so an attack is only meaningful if it improves on it.
- Test two further defenses:
  - rate limiting, modeled both per enrolled template and per attacker account;
  - detection of anomalous query sequences, e.g., flagging series of highly similar consecutive probes.

Evaluation.

- Metrics: attack success as a function of the number of queries, at two levels: (i) acceptance, i.e., the forged template is accepted at the system's operating threshold; (ii) identity recovery, i.e., the forged template matches other images of the target, not used for enrollment, at a stricter threshold (e.g., FMR = 0.01%). Show the zero-effort baseline in every plot, and report the median and 90th percentile of the number of queries.
- Data split: nothing is trained. Set all thresholds on LFW identities that are disjoint from both the targets and the starting pool. State the training data of the pretrained matcher and whether it may contain LFW identities.
- Main curves: attack success versus number of queries; for the milestone, also the genuine/impostor score distributions and the DET curve of the baseline verifier.

Reference result. No published result exists for this exact quantity. Use (i) the published verification accuracy of your pretrained matcher on LFW as the baseline reference, and (ii) a published query-based template-reconstruction attack for a qualitative comparison of query counts.

Beyond the course. Attacks on protected (cancelable) templates, image-space reconstruction, and auditing of deletion/unlearning claims.

B2. New Attack or New Dataset? Factorized Generalization in Face PAD

Problem. Presentation attack detection (PAD) papers report "generalization" as a single number. Unseen attack types and unseen capture conditions are different problems, and they may respond differently to the same method.

Question. How large is each failure when measured separately, and does model adaptation help or hurt each one?

Scope.

- Use CelebA-Spoof; a subset is sufficient.
- Use a pretrained foundation backbone: DINOv2 with a ViT-B/14, or CLIP (Contrastive Language–Image Pre-training) with a ViT-B/16 (Vision Transformer, base size, with 14×14 and 16×16 patches, respectively).
- Compare two adaptation regimes, both of which keep the pretrained weights frozen: a linear probe, which trains only a linear classifier on the backbone features, and low-rank adaptation (LoRA), which also trains small low-rank matrices added to the backbone.
- Evaluate under two protocols: leave-one-attack-type-out and leave-one-capture-condition-out.
- Hold out attack types at two levels: a single spoof type (e.g., Phone) and a whole attack family (print, paper cut, replay or 3D mask). When a single type is held out, similar types of the same family remain in training, so the held-out attack is not truly new; compare the two levels.
- Account for a property of CelebA-Spoof: all bona fide images come from CelebA (web photographs), and the capture-condition labels (illumination, environment) exist only for attack images. Leave-one-capture-condition-out therefore changes only the attack distribution. Report it as a shift on the attack side (APCER), and state that BPCER is not tested under unseen conditions.
- Check whether the detector relies on recapture cues that separate web photographs from recaptured images rather than on the attack itself: train a simple classifier on low-level image statistics only (e.g., resolution, JPEG quality, frequency spectrum). If it also separates the two classes almost perfectly, the task contains a shortcut, and your results must be read with this in mind.

Evaluation.

- Metrics, for each axis separately and never averaged across axes: the detection equal error rate (D-EER), and the bona fide presentation classification error rate (BPCER) at attack presentation classification error rate (APCER) $\in \{1\%, 5\%, 10\%\}$.
- Thresholds: report BPCER at the fixed APCER values with the threshold set on the test attacks, as in the literature. In addition, report the APCER and BPCER reached with a threshold chosen on validation data, which contains no held-out attack type or condition.
- Data split: attack images are produced from the same subjects' bona fide images, so all images of one subject must be in the same split.
- Main curve: DET (APCER vs. BPCER).

Reference result. A published result that uses a foundation-model backbone for face PAD under an unseen-attack protocol, and/or published intra-dataset results on CelebA-Spoof.

Beyond the course. Additional datasets, a sensor axis, and continual learning in which each increment is a new attack type.

B3. Where Do Audio Deepfake Detectors Break? Generator × Channel × Language

Problem. Audio deepfake detectors that are near-perfect on benchmarks fail on real-world audio. The possible causes are unseen generators, acoustic channels and languages, and they are usually studied one at a time.

Question. How much does each factor contribute to the failure, and do the factors compound?

Scope.

- Build a detector with a frozen self-supervised front-end (wav2vec 2.0 or WavLM) and a light back-end. Resample all audio to 16 kHz, normalize loudness and use a fixed input length.
- Train on the logical access (LA) subset of ASVspoof 2019 (Automatic Speaker Verification Spoofing and Countermeasures Challenge).
- Evaluate on three datasets:
  - the ASVspoof 2019 LA evaluation set, as the in-domain reference (11 of its 13 attacks are not seen in training);
  - In-the-Wild, as the real-world reference: the drop in performance that the factors below are meant to explain;
  - MLAAD (Multi-Language Audio Anti-Spoofing Dataset), using M-AILABS for bona fide speech.
- Separate language from corpus: ASVspoof is English only, while MLAAD and M-AILABS form a different corpus. Build an English MLAAD/M-AILABS condition, so that ASVspoof-English vs. MLAAD-English measures the corpus effect and MLAAD-English vs. other languages measures the language effect. Limit the language axis to the eight languages for which M-AILABS provides bona fide speech (English, French, German, Italian, Polish, Russian, Spanish, Ukrainian).
- Keep the factors apart: measure the generator axis by comparing TTS systems within MLAAD-English, and measure the language axis with TTS systems that MLAAD provides in several of these languages, so that the generator is held fixed while the language changes.
- Add channel variation through codec and room-impulse simulation, applied identically to bona fide and spoofed audio.

Evaluation.

- Metrics: EER per condition with bootstrap confidence intervals. Report the interaction effects between the factors explicitly, computed on the probit scale $\Phi^{-1}(\text{EER})$, the scale of the DET plot, rather than on raw EER. The attacks are injected into the system rather than presented to a sensor, so use bona fide/spoof terminology and EER instead of APCER/BPCER.
- Data split: use the official ASVspoof 2019 LA train/development/evaluation partitions, which are speaker-disjoint, and select models on the development set only. In-the-Wild, MLAAD and M-AILABS are used for evaluation only. State the corpora on which the self-supervised front-end was pretrained.
- Main curve: DET.

Reference result. A published EER of a detector with a self-supervised front-end on the ASVspoof 2019 LA evaluation set and on In-the-Wild.

Beyond the course. A full factorial on larger corpora, and open-set attribution of the generator.

B4. How Does Open-Set Error Grow with the Gallery?

Problem. Closed-set rank-1 accuracy is saturated on many benchmarks, while open-set error on the same data remains high. Real identification systems are open-set: the probe may not be in the gallery.

Question. How does open-set error scale with gallery size and with the source of the distractors?

Scope.

- Use pretrained matchers without retraining:
  - for face, ArcFace;
  - for speaker, an ECAPA-TDNN trained on VoxCeleb2 only, so that all VoxCeleb1 speakers are unseen.
- Build each open-set test from three disjoint groups of identities:
  - enrolled identities, each with a mated probe;
  - distractors: additional gallery identities without a mated probe;
  - non-mated probes: identities that are not in the gallery at all.
- Vary the gallery size over two orders of magnitude by adding distractors.
- Draw the distractors from two sources: the same corpus as the enrolled identities, and a different, identity-disjoint corpus. Take the non-mated probes from the same corpus as the enrolled identities in both cases.
- For face, pretrained matchers are usually trained on large web-collected datasets of public figures, which may contain identities of public face corpora. Choose corpora that reduce this risk, and state the remaining risk as a limitation.
- Remove identities that occur in both corpora: drop any distractor whose best match to an enrolled identity or to a non-mated probe identity exceeds a very strict threshold (e.g., FMR = 0.001%), and report how many were removed. Otherwise a "non-mated" search may in fact have a mate in the gallery.
- For speaker, VoxCeleb2 cannot serve as a distractor corpus, because it is the matcher's training data. Use another corpus (e.g., LibriSpeech or CN-Celeb), and note that it also differs in language, recording conditions or both.

Evaluation.

- Metrics: the false negative identification rate (FNIR) at false positive identification rate (FPIR) $\in \{1\%, 0.1\%\}$, with rank-1 accuracy alongside; fit the scaling behavior with gallery size. As a reference for the fit, use the prediction from verification scores: if non-mated scores were independent, a gallery of N identities would give FPIR $\approx N \times$ FMR, so FNIR at gallery size N is expected to be close to the FNMR at FMR = FPIR/N.
- Thresholds: the FPIR thresholds may be set on the non-mated test searches, separately for each gallery size, as in large-scale evaluations; state that you did so.
- Data split: nothing is trained. For speaker, the VoxCeleb2-only matcher ensures that no test identity was seen in training; for face, reduce and report the overlap risk as described in the scope.
- Main curve: a DET curve of FNIR vs. FPIR for at least one gallery size; for the milestone, also the mated and non-mated search score distributions.

Reference result. The published verification EER or closed-set rank-1 accuracy of your matchers, as the baseline reference. No published gallery-scaling result is expected to exist.

Beyond the course. More matchers and modalities, and a proposed open-world reporting standard.